# Jordi Laguarta Soler

PhD candidate, MIT EECS · Earth Intelligence Lab · Cambridge, MA

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I build multimodal ML systems end to end for domains where expert labels are scarce: foundation models as annotators, self-supervised pretraining, explainable predictions, and national-scale inference evaluated under distribution shift. At MIT and Microsoft Research, I have applied them to high-stakes problems in health and food security: screening for COVID-19 and Alzheimer's from coughs and speech, and mapping crops across entire countries. I lead research on evaluation science: whether geospatial foundation-model benchmarks detect real differences between models or just seed noise. I also set up clinical data collaborations and mentor student researchers. PhD expected February 2027. Looking for research and applied scientist roles; open to forward-deployed and research-product work.

## EXPERIENCE

**MIT Earth Intelligence Lab** · PhD researcher (advisor: Prof. Sherrie Wang) Sep 2022 – present

- **National crop-type map of India** (first author, manuscript in preparation). Built the label pipeline: Gemini 3.1 Pro produced 728,490 crop labels from street-level images across all 737 agricultural districts (90.2% agreement with agronomists, six classes). A classifier on AlphaEarth satellite embeddings maps every 10 m pixel of India in under 4 hours: 77.1% overall accuracy (95% CI 76.1–78.1) on 17,247 expert-labelled test points; district crop areas correlate with official statistics at r = 0.60–0.96 across crops.
- The embedding classifier maps India in under 4 hours on Earth Engine; a time-series CNN reaches 85.1% on the same test set but would need an estimated 1–2 years of national inference.
- **Thailand crop-type map** (first author, AAAI 2024). Trained CNNs to label crops in Google Street View (~81,000 labels) for a national 10 m map of four crops at 93% accuracy; GPT-4V zero-shot labels gave 92%, within one point of expert-trained labels.
- **Earth-observation foundation models.** Lead EOArena, an analysis of which Earth-observation benchmarks can tell foundation models apart. Benchmark foundation models (Panopticon, AnySat, DINOv3, DOFA, ScaleMAE) across tasks and training-set sizes.
- Mentored three student researchers, about three years of supervision in total.

**Microsoft Research** · Research intern, Seattle May 2024 – Aug 2024

- Developed the architecture of a multimodal satellite foundation model for long time series (Sentinel-1, Sentinel-2, Dynamic World). Processed 20 TB of imagery; trained and evaluated the model with distributed multi-GPU training for the AI for Agriculture and AI for Industry teams.

**EY-Parthenon** · Strategy consulting intern, Geneva Sep 2021 – Jan 2022

- Strategy work on technology programmes for consumer-goods, retail and agribusiness clients.

**MIT Auto-ID Lab** · Research assistant (supervisor: Dr. Brian Subirana) Feb 2020 – Sep 2021

- First author on a COVID-19 screening model from phone-recorded forced coughs: AUC 0.97 (98.5% sensitivity, 94.2% specificity) on the official-test subgroup of 5,320 subjects (IEEE OJEMB 2020; 715 citations). Helped collect 70,000+ recordings. A 2024 independent study (Coppock et al., *Nature Machine Intelligence*) found that recruitment bias in crowdsourced audio can inflate such results.
- First author on Alzheimer's detection from spontaneous speech with the same architecture: 93.8% accuracy on ADReSS from raw audio (n = 156; Frontiers in Computer Science 2021).
- Set up clinical data collaborations with the Catalan health system and Universidad de Guadalajara.

**Earlier** Sep 2017 – Jan 2020

- Data scientist, Nutrino Health (acquired by Medtronic, 2018): improved a food-image classifier with weakly labelled social-media data.
- Lead data scientist, ADLA (student start-up): led the team building a computer-vision recommendation chatbot.
- Lead software engineer, InAGlobe Education: co-built a platform matching engineering students with NGO technical problems.

## PUBLICATIONS

1. **J. Laguarta Soler**, S. Cheng, E. Liao, S. Wang. An open 10-meter multi-crop map of India from street-level imagery and vision-language models. *Manuscript in preparation*, 2026.
2. **J. Laguarta Soler**, T. Friedel, S. Wang. Combining deep learning and street view imagery to map smallholder crop types. *Proc. AAAI* 38(20):22202–22212, 2024.
3. **J. Laguarta**, F. Hueto, B. Subirana. COVID-19 artificial intelligence diagnosis using only cough recordings. *IEEE Open Journal of Engineering in Medicine and Biology* 1:275–281, 2020.
4. **J. Laguarta**, B. Subirana. Longitudinal speech biomarkers for automated Alzheimer's detection. *Frontiers in Computer Science* 3:624694, 2021.
5. B. Subirana, F. Hueto, P. Rajasekaran, **J. Laguarta**, et al. Hi Sigma, do I have the coronavirus? Call for a new artificial intelligence approach to support health care professionals dealing with the COVID-19 pandemic. *arXiv:2004.06510*, 2020.

Research featured in BBC News, The Economist, World Economic Forum and MIT News (2020, 2024).

## EDUCATION

**MIT** · PhD, Electrical Engineering and Computer Science — Sep 2022 – Feb 2027 (expected)

**Cornell Tech** · MEng, Computer Science · merit scholarship — Sep 2020 – May 2022 (leave, fall 2021)

**Imperial College London** · BEng, Electronic and Information Engineering — Oct 2016 – May 2019

## SKILLS

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| Machine learning | Multimodal learning, vision-language models, foundation-model pretraining and evaluation, satellite time series, weak supervision, audio and speech |
| Tools | Python, PyTorch, TensorFlow, Google Earth Engine, Vertex AI and Google Cloud, distributed multi-GPU training, SLURM, SQL, C/C++ |
| Languages | Catalan, Spanish, English, French |